UNIVERSITY OF KWAZULU-NATAL INYUVESI YAKWAZULU-NATALI

UNIVERSITY OF KWAZULU-NATAL INYUVESI YAKWAZULU-NATALI

ISTN7MP MOBILE AND PERSONAL INFORMATION SYSTEMS HONOURS MODULE – SEMESTER 1 - 2026

PROF PRABHAKAR RONTALA SUBRAMANIAM

MOBILE AND PERSONAL INFORMATION SYSTEMS

- PART I : MOBILE COMPUTING
- PART II : MOBILE AND PERSONAL INFORMATION SYSTEMS
- PART III : ANDROID MOBILE APP DEVELOPMENT

PART II

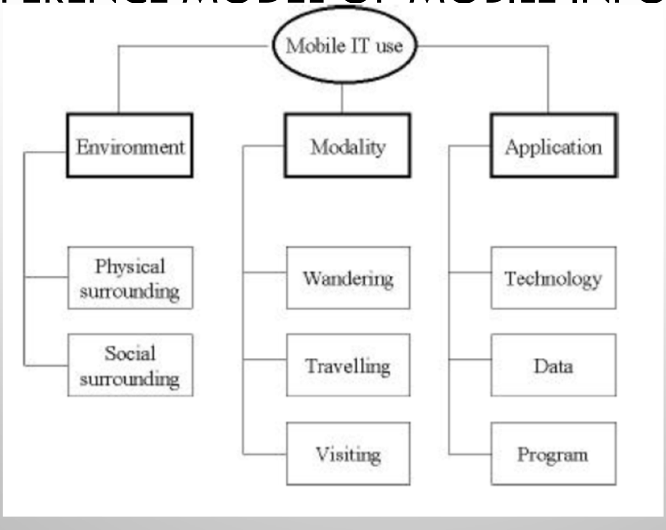
MOBILE AND PERSONAL INFORMATION SYSTEMS

MOBILE INFORMATICS

MOBILE INFORMATICS

- USING IT IN STATIONARY SETTINGS DIFFERS FROM USING IT IN MOBILE SETTINGS
- A REFERENCE MODEL TO UNDERSTAND HOW PEOPLE USE IT IN MOBILE SETTINGS IS CALLED MOBILE INFORMATICS
- KRISTOFFERSEN & LJUNGBERG (2000) HAS PROPOSED A REFERENCE MODEL FOR MOBILE IT USE
- THREE CORE AREAS OF CONCERN IN THE REFERENCE MODEL ARE ENVIRONMENT, MODALITY AND APPLICATION

REFERENCE MODEL OF MOBILE INFORMATICS



MOBILE IT USE - ENVIRONMENT

- ENVIRONMENT OF MOBILE IT USE REFERS TO THE IT ENABLED SURROUNDING OF THE MOBILE DEVICE AND THE SYSTEM BEING USED
  - THE PHYSICAL SURROUNDING/ENVIRONMENT
  - THE SOCIAL SURROUNDING/ENVIRONMENT

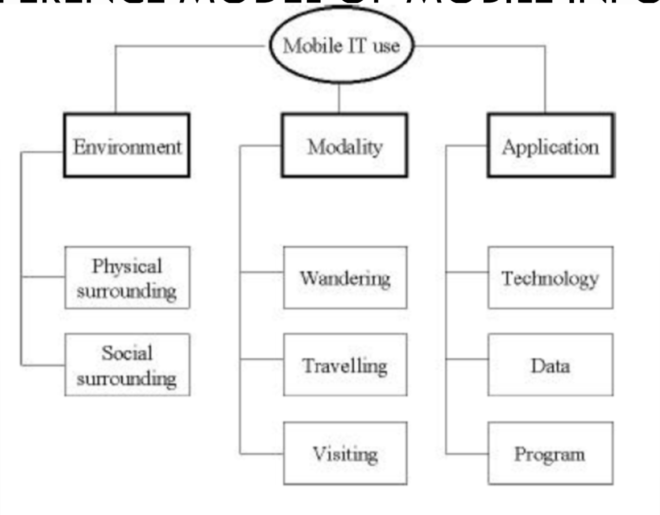
MOBILE IT USE – ENVIRONMENT (CONTD…)

- THE PHYSICAL ENVIRONMENT
  - WITH RESPECT TO PERCEPTION AND SENSES THAN CAN BE USED IN ICT
    - PHOTONS – LIGHT -> SEEING
    - MECHANIC VIBRATIONS – SOUND -> HEARING
    - MATERIALS – TOUCH -> FEEL
    - OTHERS (SMELL AND TASTE) – E-NOSE, TASTE -> NOT WIDELY USED IN ICT
  - MOBILE SYSTEM IS CAPABLE OF CONNECTING THE BASE STATION TO OTHER NODES IN OTHER LOCATIONS
    - PERSONAL AREA NETWORK (PAN)
    - IMMEDIATE ENVIRONMENT
    - INSTANT PARTNERS
    - RADIO ACCESSES
    - INTERCONNECTIVITY
    - CYBERWORLD

MOBILE IT USE – ENVIRONMENT (CONTD…)

- THE SOCIAL ENVIRONMENT
  - UNLIKE PHYSICAL ENVIRONMENT, SOCIAL ENVIRONMENT IS NOT DIRECTLY OBSERVABLE INCLUDING POWER STRUCTURES, TEAM ORGANIZATION, ETC.
  - WITH RESPECT TO HUMAN UNLIKE OLDER DAYS THERE IS MUCH MORE TO SOCIALITY AND SOCIALIZING THAN WORK
  - MOBILE TECHNOLOGIES HAVE PROVEN SIGNIFICANT IN A WIDE RANGE OF SOCIAL PURPOSES
  - CONCEPT OF A FRIEND AND FAMILY HAS BEEN SPECULATED TO GO UNDER CHANGE THESE DAYS

REFERENCE MODEL OF MOBILE INFORMATICS



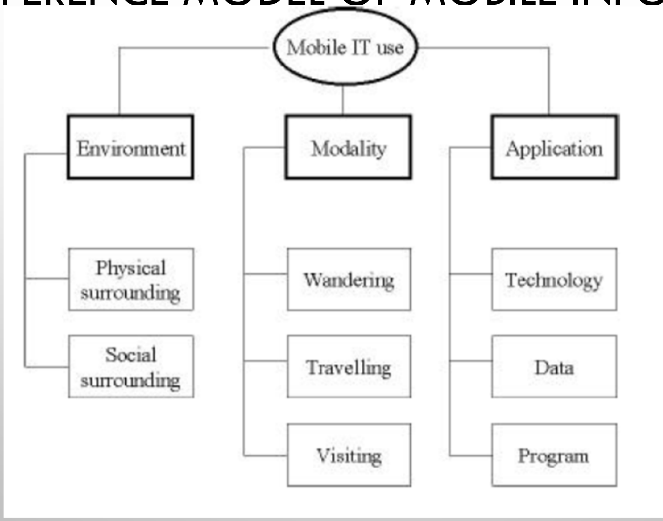
MOBILE IT USE - MODALITY

- MODALITY REFERS TO THE MODE OF MOBILITY
  - WANDERING
  - TRAVELLING
  - VISITING

MOBILE IT USE – MODALITY (CONTD…)

- WANDERING - ACTIVITY CHARACTERIZED BY EXTENSIVE LOCAL MOBILITY. EXAMPLE, ICS SUPPORT PERSON SPENDING CONSIDERABLE PART OF HIS/HER WORK TIME WANDERING AROUND THE BUILDING
- TRAVELLING – ACTIVITY THAT TAKES PLACE WHILE ON MOVE IN A VEHICLE.
- VISITING – ACTIVITY THAT HAPPENS IN ONE PLACE FOR A COHERENT BUT TEMPORAL PERIOD OF TIME. FOR EXAMPLE, A CONSULTANT WHO STAYS IN A CUSTOMER ORGANIZATION FOR A COUPLE OF DAYS, BEFORE LEAVING TO THE NEXT STATION
- ACCORDING TO HJELM (2000) “LAPTOPS ARE NOT MOBILE”
- ACCORDING TO KRISTOFFERSEN (1998) ANOTHER CLASSIFICATION BETWEEN MOBILITY – LOCAL, REGIONAL AND GLOBAL MOBILITY

REFERENCE MODEL OF MOBILE INFORMATICS



MOBILE IT USE – APPLICATION

- APPLICATION IN THE CONTEXT OF MOBILE IT USE ENCOMPASSES A DISCUSSION ON
  - TECHNOLOGY (DISCUSSED EARLIER)
  - DATA (TO BE DISCUSSED LATER)
  - PROGRAM

MOBILE INFORMATION SYSTEMS

MOBILE INFORMATION SYSTEMS – COMPUTING PARADIGMS

- ACCORDING TO DIFFERENT AUTHORS DIFFERENT FORMS OF MOBILE COMPUTING
  - UBIQUITOUS COMPUTING (WEISER, 1993)
  - NOMADIC COMPUTING (LI & LEUNG, 1997)
  - INVISIBLE COMPUTING (NORMAN, 1999)

MOBILE INFORMATION SYSTEMS - COMPONENTS

- MOBILE INFORMATION SYSTEMS CAN BE DESCRIBED AS SYSTEMS INVOLVING:
  - MOBILE DEVICES
  - USERS
  - WIRELESS AND MOBILE NETWORKS
  - MOBILE APPLICATIONS
  - DATABASES AND
  - MIDDLEWARE
- ADVANCES IN EACH ONE OF THESE AREAS INFLUENCE ALL MOBILE AND WIRELESS INFORMATION SYSTEMS. FOR EXAMPLE, FASTER DATABASES COULD REDUCE THE END-TO-END DELAYS (LATENCY) FOR MOBILE APPLICATIONS

MOBILE INFORMATION SYSTEMS – WORK FORCE

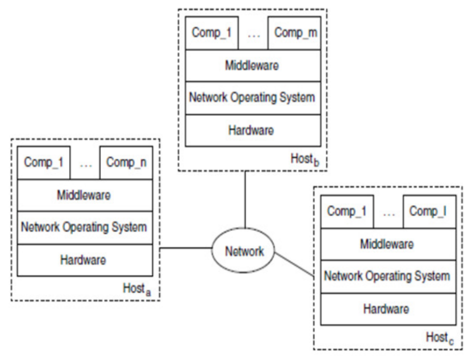
- DEVELOPMENT OF MOBILE INFORMATION SYSTEMS IS STILL A TECHNOLOGY DRIVEN EFFORT – AVAILABILITY OF POWER, PROCESSING POWER, MEMORY (PRIMARY AND SECONDARY), AND DATA COMMUNICATIONS, ETC.
- TERMS "MOBILE INFORMATION SYSTEMS" AND "MOBILE WORK" ARE USED MORE LIKE SYNONYMS FOR PORTABLE OFFICES AND TELEWORK
- MOBILE EXECUTIVES REFERS TO "TRAVELLING EXECUTIVES WHEN THEY ARE PERFORMING THEIR DUTIES OUTSIDE THEIR PERMANENT OFFICE BUILDING"
- MOBILE WORKER REFERS TO "A WORKER WHO WORKS AT A CUSTOMER'S SITE OR SOME OTHER LOCATION CHOSEN BY HIM/HER"
- BYOD TREND DRIVES THE WORK FORCE BEHIND MOBILE INFORMATION SYSTEMS AND IT'S A SEPARATE STUDY BY ITS OWN

MOBILE INFORMATION SYSTEMS – CONTEXT AWARENESS

- MOBILE INFORMATION SYSTEMS AIMS TO CONTEXT AWARENESS TO ADAPT FUNCTIONALITY
- CONTEXT AWARENESS ENABLES MOBILITY
- CONTEXT AWARENESS REFERS TO LOCATION IN GENERAL BUT ALSO INCLUDES
  - IDENTITY
  - SPATIAL INFORMATION (LOCATION, ORIENTATION, SPEED, ACCELERATION)
  - TEMPORAL INFORMATION (TIME OF THE DAY, DATE, SEASON OF THE YEAR)
  - ENVIRONMENTAL INFORMATION (TEMPERATURE, AIR QUALITY, LIGHT OR NOISE LEVEL)
  - SOCIAL SITUATION (WHO YOU ARE WITH, PEOPLE THAT ARE NEARBY)
  - RESOURCES THAT ARE NEARBY (ACCESSIBLE DEVICE AND HOSTS)
  - AVAILABILITY OF RESOURCES (BATTERY, DISPLAY, NETWORK, BANDWIDTH)
  - PHYSIOLOGICAL MEASUREMENTS (BLOOD PRESSURE, HEART RATE, RESPIRATION RATE, ETC)
  - ACTIVITY (TALKING, READING, WALKING, RUNNING)

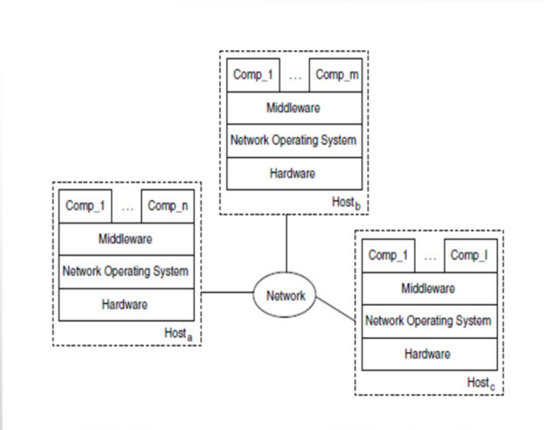
DISTRIBUTED SYSTEMS - CHARACTERIZATION

- DISTRIBUTED SYSTEM CONSISTS OF COLLECTION OF COMPONENTS DISTRIBUTED OVER VARIOUS COMPUTERS (ALSO CALLED HOSTS) CONNECTED VIA A COMPUTER NETWORK
- COMPONENTS IN A DISTRIBUTED SYSTEM ARE **INDEPENDENT, AUTONOMOUS, AND NETWORKED COMPUTING ELEMENTS**—SOFTWARE PROCESSES, HARDWARE DEVICES, OR SERVICES—THAT WORK TOGETHER TO APPEAR AS A SINGLE, COHERENT SYSTEM.
- COMPONENTS INTERACT WITH EACH OTHER TO EXCHANGE DATA AND SERVICES TO ACHIEVE INTENDED TASK



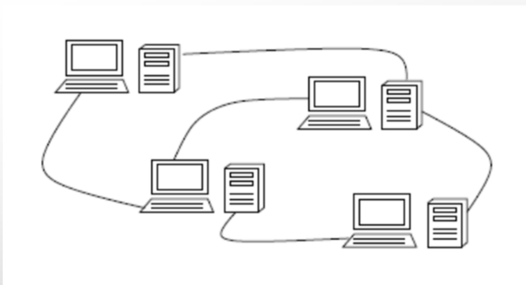
DISTRIBUTED SYSTEMS – CHARACTERIZATION (CONTD…)

- INTERACTION BUILT ON TOP OF NETWORK OPERATING SYSTEM MAY BE TOO COMPLEX FOR MANY APPLICATION DEVELOPERS
- A MIDDLEWARE IS LAYERED BETWEEN DISTRIBUTED SYSTEM COMPONENTS AND NETWORK OPERATING SYSTEM COMPONENTS TO FACILITATE INTERACTION



TRADITIONAL DISTRIBUTED SYSTEMS

- NON-FUNCTIONAL REQUIREMENTS
  - SCALABILITY
  - OPENNESS
  - HETEROGENEITY
  - FAULT-TOLERANCE
  - RESOURCE SHARING



TRADITIONAL DISTRIBUTED SYSTEMS (CONTD…)

- NON-FUNCTIONAL REQUIREMENTS
  - SCALABILITY – ABILITY TO REGULATE LOAD (HIGHER OR LOWER) IN TERMS OF NUMBER OF USERS, NUMBER OF TRANSACTIONS, ETC
  - OPENNESS – POSSIBILITY TO EXTEND AND MODIFY THE SYSTEM EASILY AS A RESULT OF CHANGING FUNCTIONAL REQUIREMENTS
  - HETEROGENEITY – INTEGRATION OF COMPONENTS DEVELOPED USING DIFFERENT PROGRAMMING LANGUAGES, FROM DIFFERENT OPERATING SYSTEMS AND RUNNING ON DIFFERENT HARDWARE PLATFORMS
  - FAULT-TOLERANCE – ABILITY TO RECOVER FROM FAULT WITHOUT TERMINATING THE SERVICES
  - RESOURCE SHARING – IN TERMS OF HARDWARE, SOFTWARE, NETWORK BANDWIDTH ETC

MOBILE DISTRIBUTED SYSTEMS - CHARACTERIZATION

- THE DEFINITION DISCUSSED ABOVE ON DISTRIBUTED SYSTEMS APPLIES TO BOTH FIXED AND MOBILE DISTRIBUTED SYSTEMS
- THE FOLLOWING THREE CONCEPTS DIFFERENTIATE BETWEEN FIXED AND MOBILE DISTRIBUTED SYSTEMS
  - TYPE OF DEVICE – FIXED OR MOBILE (SELF EXPLANATORY AND SELF STUDY)
  - TYPE OF NETWORK – PERMANENT OR INTERMITTENT (DISCUSSED EARLIER)
  - TYPE OF EXECUTION CONTEXT – STATIC OR DYNAMIC

TYPE OF EXECUTION CONTEXT

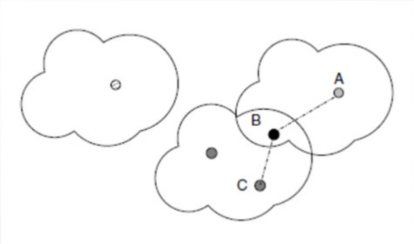
- CONTEXT OF EXECUTION REFERS TO EVERYTHING THAT INFLUENCE THE BEHAVIOR OF THE APPLICATION
- CONTEXT INCLUDES RESOURCES INTERNAL AND EXTERNAL TO THE DEVICE
  - INTERNAL RESOURCES INCLUDES – AMOUNT OF MEMORY, SCREEN SIZE, POWER, ETC.
  - EXTERNAL RESOURCES INCLUDES – BANDWIDTH, QUALITY OF NETWORK CONNECTION, LOCATION OR HOSTS IN THE PROXIMITY (SERVICE)
- IN FIXED DISTRIBUTED SYSTEMS THE RESOURCES ARE ALL STATIC
- IN MOBILE DISTRIBUTED SYSTEMS THE RESOURCES ARE ALL DYNAMIC

NOMADIC DISTRIBUTED SYSTEMS

- INTERMEDIATE SYSTEMS – A COMPROMISE BETWEEN TOTALLY FIXED AND TOTALLY MOBILE DISTRIBUTED SYSTEMS
- NOMADIC SYSTEMS COMPOSED OF A SET OF MOBILE DEVICES AND A CORE INFRASTRUCTURE WITH FIXED AND WIRED NODES
- IN NOMADIC SYSTEMS THE MOBILE DEVICES MOVE FROM LOCATION TO LOCATION, WHILE MAINTAINING A CONNECTION TO THE FIXED NETWORK
- SERVICES ARE MAINLY PROVIDED BY THE CORE INFRASTRUCTURE TO THE MOBILE NODES

AD-HOC MOBILE DISTRIBUTED SYSTEMS

- AD-HOC MOBILE DISTRIBUTED SYSTEMS CONSISTS OF SET OF MOBILE HOSTS, CONNECTED TO THE NETWORK THROUGH HIGH-VARIABLE QUALITY LINKS, AND EXECUTING IN AN EXTREMELY DYNAMIC ENVIRONMENT
- THEY DIFFER WITH NOMADIC DISTRIBUTED SYSTEMS BY NOT DEPENDING UPON FIXED INFRASTRUCTURE
- MOBILE HOSTS CAN ISOLATE THEMSELVES AND GROUPS MAY EVOLVE INDEPENDENTLY



MOBILE MIDDLEWARE

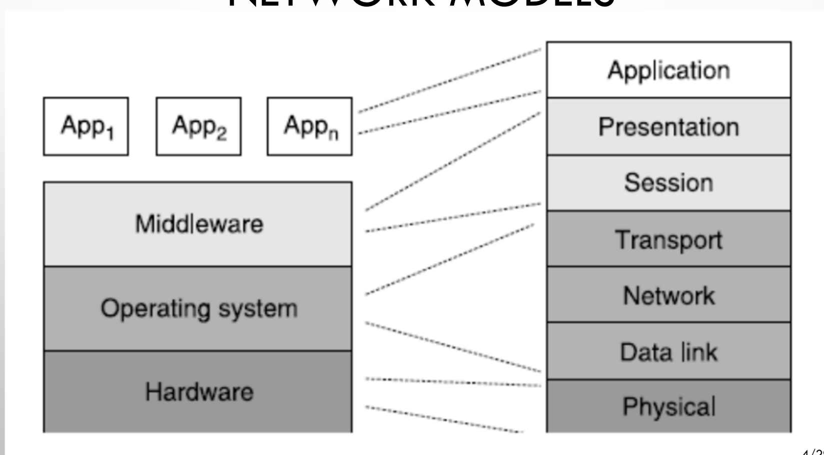
- MOBILE MIDDLEWARE IS SOFTWARE THAT CONNECTS DISPARATE MOBILE APPLICATIONS, PROGRAMS AND SYSTEMS
- MOBILE MIDDLEWARE ESSENTIALLY HIDES COMPLEXITIES OF WORKING IN MOBILE ENVIRONMENTS, ALLOWING FOR SMOOTHER DEVICE-TO-DEVICE INTERACTION, MOBILE COMPUTING INTEGRATION AND MOBILE APPLICATION DEVELOPMENT
- LIKE OTHER MIDDLEWARE MOBILE MIDDLEWARE ALSO PROVIDES MESSAGING SERVICES TO ENABLE COMMUNICATION BETWEEN DIFFERENT APPLICATIONS

MIDDLEWARE – A REFERENCE MODEL

- OPEN SYSTEMS INTERCONNECTION (OSI) REFERENCE MODEL ARCHITECTURE DIVIDES NETWORK PROTOCOLS INTO SEVEN LAYERS

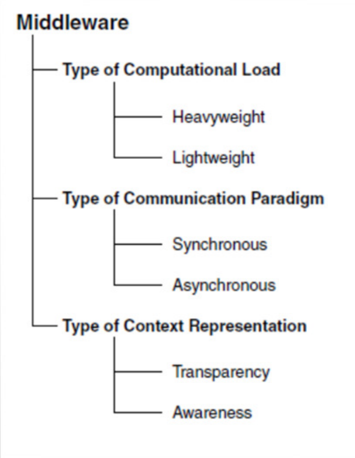
7. Application	• Provides a user interface
6. Presentation	• Presents Data • Handles encryption and decryption
5. Session	• Maintains distinction between data of separate applications • Provides dialog control between hosts
4. Transport	• Provides End-to-End connections • Provides reliable or unreliable delivery and flow control
3. Network	• Provides Logical Addressing • Provides Path determination using logical addressing
2. Data Link	• Provides media access and physical addressing
1. Physical	• Converts digital data so that it can be sent over the physical medium • Moves data between hosts

RELATIONSHIP BETWEEN SOFTWARE MODELS AND NETWORK MODELS



MOBILE MIDDLEWARE - CHARACTERIZATION

- TYPE OF COMPUTATION LOAD
  - HEAVY WEIGHT – REQUIRES LARGE AMOUNT OF RESOURCES
  - LIGHT WEIGHT – REQUIRES LESS RESOURCES
- TYPE OF COMMUNICATION PARADIGM
  - SYNCHRONOUS – BOTH CLIENT AND SERVER WORK AT THE SAME TIME
  - ASYNCHRONOUS – CAN WORK AT DIFFERENT TIME
- TYPES OF CONTEXT REPRESENTATION
  - AWARENESS – INFORMATION ABOUT THE EXECUTION CONTEXT IS FED TO THE APPLICATIONS
  - TRANSPARENCY – INFORMATION ABOUT THE EXECUTION CONTEXT IS HIDDEN INSIDE THE MIDDLEWARE



MOBILE INFORMATION SYSTEMS - SECURITY

- SECURITY ISSUES
  - CONFIDENTIALITY – HIDING/SECURING INFORMATION FROM OTHER USERS
  - AUTHENTICATION – MOBILE USERS MUST BE ABLE TO TRUST THE IDENTITY CLAIMED
  - INTEGRITY – DATA/TRANSACTIONS SHOULD NOT BE MODIFIED BY OTHERS
  - AUTHORIZATION – PARTIES INVOLVED IN TRANSACTION BE ABLE TO VERIFY EACH OTHER
  - NON-REPUDIATION – USER SHOULD NOT CLAIM THAT THE TRANSACTION WAS MADE WITHOUT HIS/HER KNOWLEDGE

MOBILE INFORMATION SYSTEMS - SECURITY

- HIGH LEVEL THREATS AND VULNERABILITIES
  - LACK OF PHYSICAL SECURITY CONTROLS
  - USE OF UNTRUSTED MOBILE DEVICES
  - USE OF UNTRUSTED NETWORKS
  - USE OF UNTRUSTED APPLICATIONS
  - INTERACTION WITH OTHER SYSTEMS
  - USE OF UNTRUSTED CONTENT
  - USE OF LOCATION SERVICES

MOBILE INFORMATION SYSTEMS - CHALLENGES

- DESIGN OF MOBILE APPLICATIONS AND SERVICES
- CONTEXT (AND LOCATION) - AWARENESS
- DEVICE AND USER INTERFACE ISSUES
- SUPPORT FOR GROUP COMMUNICATIONS
- RELIABLE COMMUNICATIONS
- INTER-WORKING AND INTEGRATION OF DIFFERENT WIRELESS TECHNOLOGIES
- INTRODUCTION OF MOBILE TECHNOLOGIES IN BUSINESS AND ORGANISATIONS

THANK YOU

END OF

MOBILE INFORMATION SYSTEMS